CAAM/STAT31470: Homework #4

Purpose: The goal of these questions is to provide you practice for calculations and reinforce concepts from class. Unless specifically told in the question, you should always show your working and justify your answers. Marks will be awarded for your reasoning in addition to the correctness of your answers. Also note that these questions will be similar to the questions on the Midterm and Final.

Content: This week we look at contour integrals in the complex plane. We practice computing them and also look at their relationships with anti-derivatives.

Homework is due: **Wednesday April 23, 11:59pm**.

Problem 4.1 (6 points)

Compute the contour integral

$$\int_C f(z)\mathrm{d}z$$

for each of the following functions and contours:

(a) $f(z) = (z+2)/z$ and C given by $z = 2e^{i\theta}$ for $0 \leq \theta \leq \pi$

(b) $f(z) = 1$ for $y < 0$ and $f(z) = 4y$ for $y > 0$, and C given by the arc joining $-1-i$ to $1+i$ along $y = x^3$

(c) $f(z) = z^m \bar{z}^n$ where m, n are integers and C is the unit circle taken counter clockwise.

Problem 4.2 (4 points)

Let C_0 and C denote the circles

$$z = z_0 + Re^{i\theta} \quad \text{and} \quad z = Re^{i\theta}$$

respectively.

(a) Show that

$$\int_{C_0} f(z - z_0)\mathrm{d}z = \int_C f(z)\mathrm{d}z$$

(b) Use part (a) to compute

$$\int_{C_0} (z - z_0)^{n-1}\mathrm{d}z \quad \text{and} \quad \int_{C_0} \frac{\mathrm{d}z}{z - z_0}$$

where n is a nonzero integer.

Problem 4.3 (4 points)

(a) Suppose that a function $f(z)$ is continuous on a smooth arc C, which has a parametric representation $z = z(t)$ for $a \leq t \leq b$. Show that if $\phi(\tau)$ for $\alpha \leq \tau \leq \beta$ is smooth with $\phi(\alpha) = a$, $\phi(\beta) = b$ and has strictly positive derivative then

$$\int_a^b f(z(t))z'(t)\mathrm{d}t = \int_\alpha^\beta f(Z(\tau))Z'(\tau)\mathrm{d}\tau \tag{1}$$

where $Z(\tau) = z(\phi(\tau))$.

(b) Argue that result also holds when C is a general contour.

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Problem 4.4 (8 points)

Let C_R be the circle with radius R oriented counterclockwise. Show that

$$\left| \int_{C_R} \frac{\mathrm{Log}(z)}{z^2} \mathrm{d}z \right| < 2\pi \frac{\pi + \ln(R)}{R} \tag{2}$$

and hence determine the limit of the contour integral as R goes to infinity.

Problem 4.5 (8 points)

Show that

$$\int_{-1}^{1} z^i \mathrm{d}z = \frac{1 + e^{-\pi}}{2}(1 - i) \tag{3}$$

where the integrand denotes the principal branch

$$z^i = \exp(i\mathrm{Log}(z)), \quad \text{for } |z| > 0,\ -\pi < \mathrm{Arg}(z) < \pi \tag{4}$$

of z^i and where the path of integration is any contour from $z = -1$ to $z = 1$ that, except for its endpoints, lies above the real axis.

End of homework